Tubular Daylight Tube System - Expanded Shop Guide

Detailed, easier-to-read companion to the 14-inch / 356 mm prototype tube fabrication package.

1	2	3	4	5
UNDERSTAND	PLAN	FABRICATE	INSPECT	VALIDATE

DOCUMENT STATUS	Prototype fabrication and optical development only. The roof, structural attachment, flashing, firestop, insulation/vapor design, and permanent building release remain project-specific.

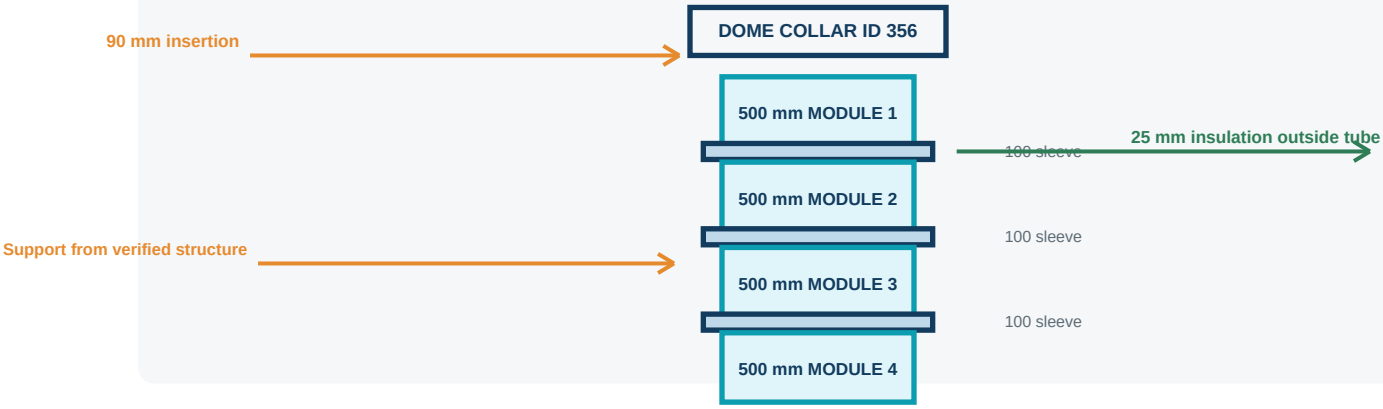
What this expanded edition adds

- Plain-language explanation of every major interface and why each controlling dimension matters.
- Simplified diagrams for the optical path, joints, supports, seals, routing, and service interfaces.
- Step-by-step fabrication and installation travelers with CHECK and HOLD gates.
- Inspection logic, common failure clues, commissioning records, and prototype release sign-off.
- The original engineering package is appended unchanged and remains the controlling dimensional source.

HOW TO USE THIS DOCUMENT	Read the explanatory supplement first. Use the original sheets for fabrication geometry and exact source values. Do not treat a plain-language explanation or example calculation as a new released design value.

1. End-to-end system map

The tube package begins at the existing dome lower collar and ends at the room-side diffuser. The baseline source configuration uses four 500 mm reflective modules, but the measured site distance controls the final total cut length.



ZONE	SOURCE CONTROL	WHY IT MATTERS
Dome collar	ID 356; tube insertion 90	Sealed upper socket; keep hardware outside bore
Reflective tube	OD 355 +/-0.5; wall 0.50 +/-0.02	High-reflectance light path; keep round, smooth, clean
Joint sleeve	100 wide; 356 bore; butt gap 0-1	External alignment and seal; no internal fasteners
Supports	<=1000 spacing; <=200 near transitions	Carry weight to verified structure; allow one axial movement point
Thermal envelope	25 mm closed-cell starting basis	Condensation/vapor control; final design is project-specific
Plenum	ID 356 x 150 high; insertion 75	Lower socket and structural room-side interface
Diffuser	rough opening Dia. 400 +/-2; clear Dia. 380	Serviceable distribution layer; verify glare/illuminance

OPTICAL RULE	Every internal step, dent, fastener head, fingerprint, or ovalized section can scatter light. The package keeps connections external so the optical bore remains smooth.

2. How to read the drawings and shop language

This page translates the recurring drawing terms into workshop meaning. It does not change any source dimension.

TERM	PLAIN-LANGUAGE MEANING
OD / ID	Outside diameter / inside diameter. The tube is OD 355 and interfaces with nominal ID 356 sockets/sleeves.
Optical bore	The clear reflective interior light path. Nothing should project into it.
Developed blank	Flat sheet size before rolling into a cylinder or sleeve.
Longitudinal lap	The lengthwise seam where the flat blank overlaps itself after rolling.
Butt joint	Two tube ends meet end-to-end rather than telescoping into each other.
Split sleeve	External cylindrical clamp around a butt joint. It aligns and supports without internal hardware.
Insert / engagement	How far the tube enters the dome collar or plenum socket.
Roundness max-min	Difference between largest and smallest measured diameters around a tube end.
End squareness	How perpendicular the cut end is to the tube axis.
NTS	Not to scale. Read the stated dimensions, not visual proportions.
HOLD / project-specific	Work may continue only after the responsible discipline closes the unresolved requirement.

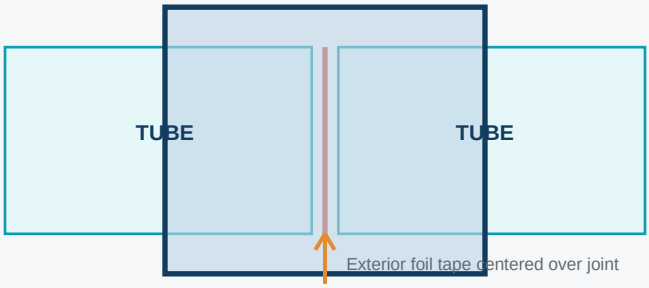
DIMENSION PRIORITY	If a sketch looks different from the printed dimension, use the printed dimension. The original engineering sheets appended at the back remain controlling.

3. Critical interface chain and tolerance checks

The most important fit chain is 356 mm socket/sleeve interfaces around a 355 mm tube. The design intentionally keeps a small assembly clearance while limiting ovality and joint steps.

FEATURE	SOURCE LIMIT	PRACTICAL CHECK
Tube OD	355.0 +/-0.5	Measure at multiple angular positions at both ends
Approx. tube ID	354.0	Reference; controlled by OD and 0.50 wall
Dome collar ID	356.0	Tube enters 90 mm; maintain 10 mm top clearance
Joint sleeve bore	356.0	Trial fit; do not tighten until tube ovalizes
Plenum socket ID	356 +/-0.5	Tube enters 75 mm; support plenum from framing
Roundness	max-min <=1.0	Reject/repair rolling distortion before assembly
End squareness	<=0.50 gap	Prevents light-scattering steps and sleeve misalignment
Inside seam step	<=0.30	Keep lap edge toward diffuser
Butt gap	0-1.0 straight; <=1.5 elbow	Check visually before exterior tape

100 mm EXTERNAL SPLIT SLEEVE - 356 BORE



BUTT GAP 0-1 mm; all hardware remains outside optical bore

DO NOT FORCE FIT	A sleeve or clamp that requires crushing the tube is not a successful fit. Restore roundness or correct the sleeve before tightening.

4. Length planning - from site measurement to cut list

Measure the intended centerline distance from the dome-collar underside to the ceiling-plenum top. The source formula adds the two socket engagements: 90 mm at the dome and 75 mm at the plenum.

STEP	ACTION	WHY
1	Measure cavity distance D	Record the actual path between interface planes.
2	Total tube cut length L = D + 165 mm	165 = 90 upper insertion + 75 lower insertion.
3	Split L into sections <=600 mm	The baseline uses 500 mm sections; shorter sections may be used for handling.
4	Locate joints away from conflicts	Leave room for sleeves, tape, hangers, framing, and elbow details.
5	Dry assemble before roof/ceiling closure	Verify length, fit, bore alignment, and service access.

Worked examples - calculation only

CASE	MEASURED D (mm)	D+165 (mm)	NOTE
Source baseline	1835	2000	Matches four 500 mm modules in the package.
Example A	1200	1365	Divide into practical sections <=600 mm; exact section plan is a shop decision.
Example B	2750	2915	Keep joints accessible and support/routing rules unchanged.

MEASURE ALONG THE INTENDED CENTERLINE	If an elbow is used, plan the actual routed centerline and the elbow fabrication geometry before cutting the straight modules.

5. Straight reflective module - fabrication steps

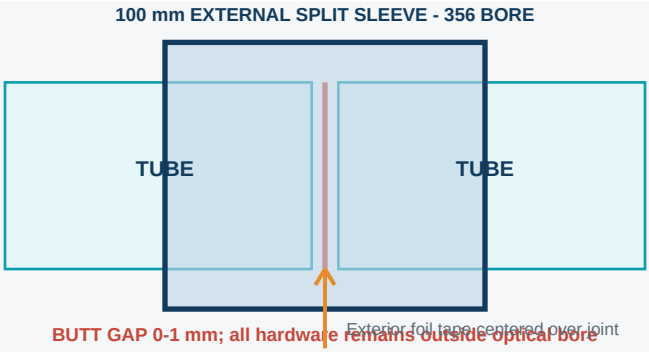
The source module is a 0.50 mm enhanced-specular aluminum cylinder. The optical face must remain protected through cutting and rolling.

OPERATION	CONTROL / EXPLANATION
1. Verify material	Enhanced-specular aluminum with documented optical performance; ordinary mill-finish aluminum is not equivalent.
2. Mark blank	1130 x module length; baseline module length 500 +/-0.5. Keep optical face protected.
3. Cut and deburr	Avoid scratching the optical surface. Remove burrs from the back edge.
4. Roll	Form to OD 355 with optical face inward. Preserve circularity.
5. Close 15 mm lap	Use compatible high-strength transfer tape in lap plus exterior foil tape as specified; no rivet heads in bore.
6. Inspect	OD, roundness, end squareness, seam step, dents, coating damage, length.
7. Orient seam	Lap edge faces toward diffuser so incoming rays see the smoother transition.

Common module rejection clues

SYMPTOM	ACTION / REASON
OD / roundness out	Re-roll on correct mandrel/form; do not solve by over-tightening sleeves.
Seam step >0.30	Rework seam closure; internal step is an optical defect.
Dent >0.30 deep	Reject or correct before assembly; dents disturb bore geometry.
Deep scratch / coating damage	Evaluate against optical material acceptance; do not hide damage with ordinary tape inside bore.

6. Butt joints and external sleeve - assembly sequence

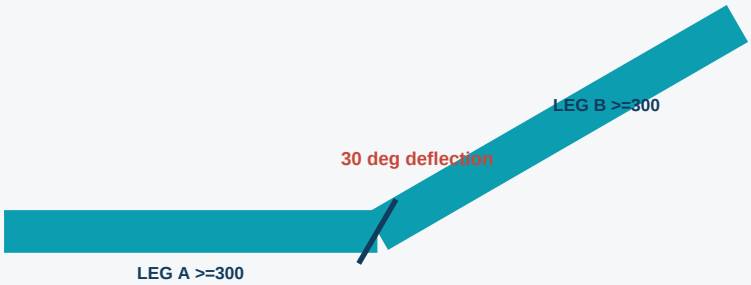


SEQ	ACTION	ACCEPTANCE LOGIC
A	Bring square tube ends together	Target butt gap 0-1 mm for straight joints.
B	Check axis alignment	Use straightedge and bore view; source verification target axis offset <=1.0.
C	Fit 356 bore split sleeve	100 mm width; developed strip 1112 x 100 x 1.0 with 8-12 mm clamp gap.
D	Tighten only enough to stop rotation	Do not crush or ovalize the reflective tube.
E	Apply exterior foil tape	150 mm wide, centered over joint as shown in source package.
F	Re-check bore	Confirm no new step, ovality, or obstruction after clamping.

WHY NO INTERNAL FASTENERS	Internal screws or rivets create bright/dark scattering features and can interrupt the smooth optical path. The source design keeps hardware outside the bore.

7. Optional 30-degree elbow - when and how to use it

The elbow is not the preferred path. Use it only where obstruction avoidance is necessary, because each bend increases reflections and reduces delivered light.



Mirrored 15 deg cuts

long-short difference = 95.1 mm

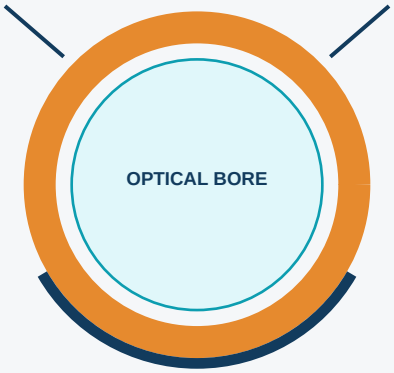
joint gap <=1.5 mm

external straps + foil tape only

ANGLE AROUND TUBE (deg)	MITER OFFSET FROM CENTERLINE (mm)
0	+47.6
30	+41.2
60	+23.8
90	0.0
120	-23.8
150	-41.2
180	-47.6

ELBOW CONTROLS	Mirrored 15-degree cuts create the 30-degree deflection. Keep each leg >=300 mm, elbow butt gap <=1.5 mm, bridge with four external 20 x 120 x 0.8 straps at 90 degrees, and seal externally with 100 mm foil tape. Support within 200 mm.

8. Structural support, insulation, and vapor continuity



25 mm closed-cell insulation

split hanger + EPDM pad

two M6 rods to verified structure

spacing <=1000 mm; <=200 mm near transitions

ITEM	SOURCE BASIS	PLAIN-LANGUAGE PURPOSE
Support spacing	<=1000 mm; within 200 mm of elbow, roof transition, plenum	Prevents long unsupported spans and concentrates support near transitions
Hanger	25 x 1.5 split band + 1 mm EPDM pad	Protects tube and spreads clamp load
Rods	Two M6 rods	Attach only to verified building structure
Thermal movement	Allow axial movement at one support in each run	Avoids locking the metal tube during temperature change
Insulation	25 mm closed-cell prototype starting basis	Final thickness and fire/smoke/condensation details are project-specific
Vapor seal	Continuous exterior layer	Seal every joint/penetration before concealment

LOAD PATH	The dome, curb cap, roof membrane, drywall, ceiling grid, and diffuser ring are not structural supports for the tube.

9. Roof adapter - dimensional interface versus building-envelope design

The source cap provides geometry for mounting the dome, but it is explicitly not a universal roof flashing. This distinction is critical for fabrication and installation responsibility.

TOPIC	EXPLANATION
Prototype interface values	500 x 500 external curb; 200 mm minimum above roof; 700 x 700 x 1.2 aluminum cap with 50 downstand; opening Dia. 370; 16 holes Dia. 5.5 on PCD Dia. 430.
Cap function	Provides a dimensional mounting surface for the dome flange and insulated tube opening.
Not defined by package	Actual roof membrane/shingle/metal-panel detailing, counterflashing, ice/water protection, curb attachment, wind resistance, drainage, fire requirements, weather seal.
Required owner	Qualified roofing and structural professionals for the specific building, climate, and code.
Shop stop point	Do not treat the generic cap as permission to open or flash a roof without the project-specific design.

HOLD - PERMANENT ROOF INSTALLATION	Close all site-specific roof, structure, fire, drainage, and weatherproofing requirements before permanent installation.

10. Ceiling plenum and diffuser - assembly and service path

The lower assembly is both an optical terminal and a service point. The plenum must be carried by framing; the decorative trim/diffuser is removable and not structural.

PART	SOURCE VALUE	ROLE
Plenum socket	ID 356 x 150 high; tube inserts 75	Sealed lower tube interface
Air-seal gasket	3 mm above ceiling	Limits room/plenum air leakage
Upper retainer	OD 410 / ID 384 x 1.0	Mechanical retention
Clear prismatic PMMA	Dia. 382 x 2.0	Upper diffuser/lens
Spacer	OD 404 / ID 384 x 15 high	Separates lens layers
Opal PMMA	Dia. 390 x 2.0	Glare diffusion; prototype transmission selection
Trim ring	OD 450 / clear Dia. 380 x 1.2	Room finish and service access
Rough opening	Dia. 400 +/-2	Ceiling coordination

DIFFUSER SELECTION	If exact transmission is unavailable, the package calls for trialing 60%, 70%, and 80% opal sheets and selecting the lowest-glare option that still meets the room illuminance target.

11. Installation traveler with stop points

#	STAGE	REQUIRED EVIDENCE	GATE
1	Site measure	Dome collar to plenum top measured; route and obstructions recorded	CHECK
2	Module plan	D+165 calculated; sections <=600; joint/hanger locations planned	CHECK
3	Fabricate tubes/sleeves	Dimensional inspections passed; optical surfaces protected	CHECK
4	Dry assemble	Overall length, alignment, joint access, diffuser service verified	CHECK
5	Roof design	Project-specific curb/attachment/flashing/roof opening approved	HOLD
6	Install upper interface	Tube insertion 90; seal/band clamp; no load on dome	CHECK
7	Install modules/supports	Joint gaps/alignment/support spacing/thermal movement checked	CHECK
8	Seal + insulate	Exterior air seals complete; insulation/vapor continuity checked	CHECK
9	Install plenum/diffuser	Insertion 75; independent framing support; removable diffuser	CHECK
10	Commission	Leak/water/support/optical/glare/thermal records complete	HOLD

DO NOT CONCEAL TOO EARLY	Keep joints, hangers, and air seals visible until inspection. Once insulation and ceiling finishes hide them, correction becomes much more difficult.

12. Inspection matrix and troubleshooting guide

CHECK	METHOD	SOURCE TARGET
Tube diameter	Measure 0/45/90/135 deg at both ends	355.0 +/-0.5; max-min <=1.0
End squareness	Flat plate + feeler gauge	<=0.50 gap
Joint alignment	Straightedge + bore inspection	Axis offset <=1.0; butt gap 0-1.0
Air leakage	Low-pressure smoke/tracer before insulation	No visible leakage at seams, joints, plenum, collar
Water control	Qualified roof spray test	No water inside curb, insulation, plenum, or room
Support	Inspect rods/brackets/clamps/thermal movement	No load on dome/flashing/drywall/diffuser
Optical throughput	Compare outdoor illuminance/flux and room-side output	Record clear-day and overcast baseline; no joint shadow bands
Glare/diffusion	Observe occupied viewpoints in low/high sun	No exposed bright tube image; adjust opal trial if needed
Thermal cycle	At least 10 hot/cold cycles with inspection	No tape release, staining, corrosion, distortion, seal movement

Failure clue -> first checks

SYMPTOM	FIRST CHECKS
Dark ring/band at joint	Gap, step, misalignment, wrinkled tape, sleeve distortion
Tube will not stay round	Rolling/tooling error or clamp overtightening
Condensation staining	Air leakage, vapor discontinuity, insufficient project-specific thermal design
Diffuser hot spot/glare	Tube alignment, diffuser trial transmission/distribution, bend count
Water at room/plenum	Roof/weather interface failure - stop and investigate before optical troubleshooting

13. Prototype record, change control, and release checklist

Use one record per fabricated installation. Attach photos, dimensional sheets, certificates, and test data. A change to geometry, materials, seals, or supports should be traceable to a new prototype/revision rather than being silently incorporated.

RECORD FIELD	ENTRY
Prototype / project ID	_____
Measured cavity D	_____ mm
Calculated cut length D+165	_____ mm
Module plan	_____
Elbow count	_____
Optical material lot/certificate	_____
Installer / fabricator	_____
Date	_____

[ ] Module OD/roundness/squareness/seam/surface records complete

[ ] Joint gaps and alignment checked before insulation

[ ] Upper and lower socket insertion/seal checked

[ ] Hangers attached only to verified structure; spacing/thermal movement checked

[ ] Exterior air seal and vapor layer continuous

[ ] Diffuser removable; ceiling opening verified

[ ] Project-specific roof/weather requirements closed

[ ] Optical/glare baseline recorded

[ ] Thermal-cycle/condensation observations recorded

[ ] All project HOLD items closed before permanent release

SOURCE PACKAGE FOLLOWS	The original 10-page engineering package is appended unchanged after this supplement. It remains the controlling source for dimensions, geometry, BOM, and stated release limitations.

COMPLETE FABRICATION DESIGN - EXCLUDING DOME

Tubular Daylight Tube System

Build-oriented drawings and specifications for the complete remaining assembly: reflective tube, modular joints, dome interface, optional elbow, curb adapter, supports, insulation and vapor control, ceiling plenum, trim ring, and diffuser.

DESIGN BASIS	CONFIGURATION	STATUS	UNITS
Existing dome collar ID 356	Four 500 mm modules	Prototype - Rev A	mm unless noted

Scope and baseline

This package starts at the existing dome's lower collar and ends at the room-side diffuser. The baseline is a straight, 2,000 mm cut-length tube stack. Because 90 mm enters the dome collar and 75 mm enters the ceiling plenum, the resulting building-cavity reach is approximately 1,835 mm. The length formula in Sheet 1 allows any other site distance to be fabricated.

PARAMETER	DESIGN VALUE	CONTROL
Tube	Dia. 355 OD x 0.50 wall	Fits existing Dia. 356 ID dome collar
Optical material	Enhanced specular aluminum	Total reflection target 98 percent
Standard module	500 long	Cut blank 1130 x 500
Joint	100 split external sleeve	Butt tube ends; no internal fasteners
Supports	At <= 1000 spacing	Two M6 rods to verified structure
Thermal envelope	25 closed-cell insulation	Continuous exterior vapor seal
Ceiling opening	Dia. 400 +/-2	Trim OD 450; clear aperture Dia. 380

CRITICAL LIMIT
The roof curb and cap shown are a generic prototype interface, not a universal roof flashing. A qualified roofing and structural professional must design the actual roof opening, curb attachment, wind resistance, drainage, flashing, fire requirements, and weather seal for the building and roof system.

Drawing index

Sheet 1 - overall assembly; Sheet 2 - straight tube module; Sheet 3 - joints and interfaces; Sheet 4 - optional elbow and supports; Sheet 5 - curb-mounted roof adapter; Sheet 6 - ceiling plenum and diffuser.

ENGINEERING DRAWING - SHEET 1 OF 6

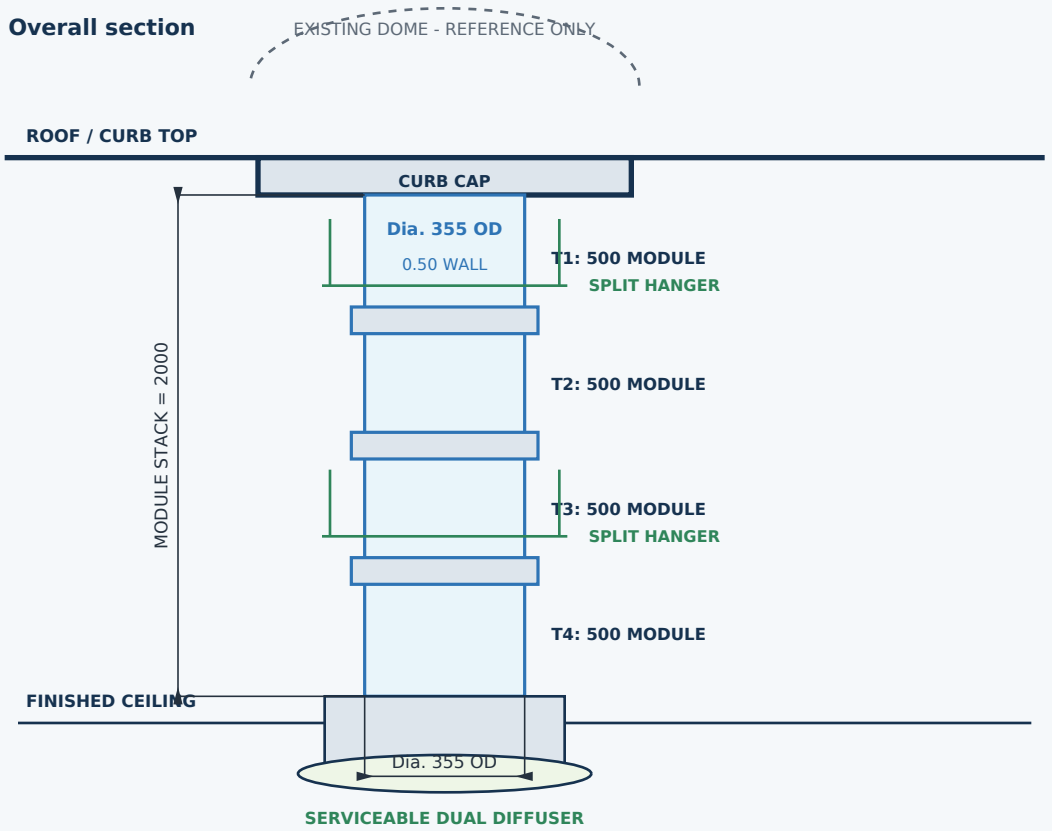
Overall assembly and controlling dimensions

COMPLETE TUBE SYSTEM - STRAIGHT BASELINE ASSEMBLY

SHEET 1 / 6

PRELIMINARY FABRICATION DESIGN - ALL DIMENSIONS mm - DRAWINGS NTS

A1 - Overall section



A2 - Controlling dimensions

Baseline: four 500 mm straight modules. The usable building-cavity reach is reduced by the 90 mm dome-collar insertion and 75 mm ceiling-plenum insertion.

Dome collar ID	**356.0**
Tube outside	**355.0 +/-0.5**
Tube inside	**approx. 354.0**
Wall	**0.50 +/-0.02**
Module length	**500.0 +/-0.5**
Stack cut length	**2000**
Installed reach	**1835**
Joint sleeve	**100 wide**
Insulation OD	**approx. 405**
Rough opening	**450 x 450 min.**

Length rule: Required total tube cut length = measured distance from dome-collar underside to plenum top + 165 mm. Divide into sections no longer than 600 mm for hand rolling and handling.

The baseline uses four 500 mm reflective modules. Use the length rule on the sheet to adapt the run to the actual roof-to-ceiling measurement.

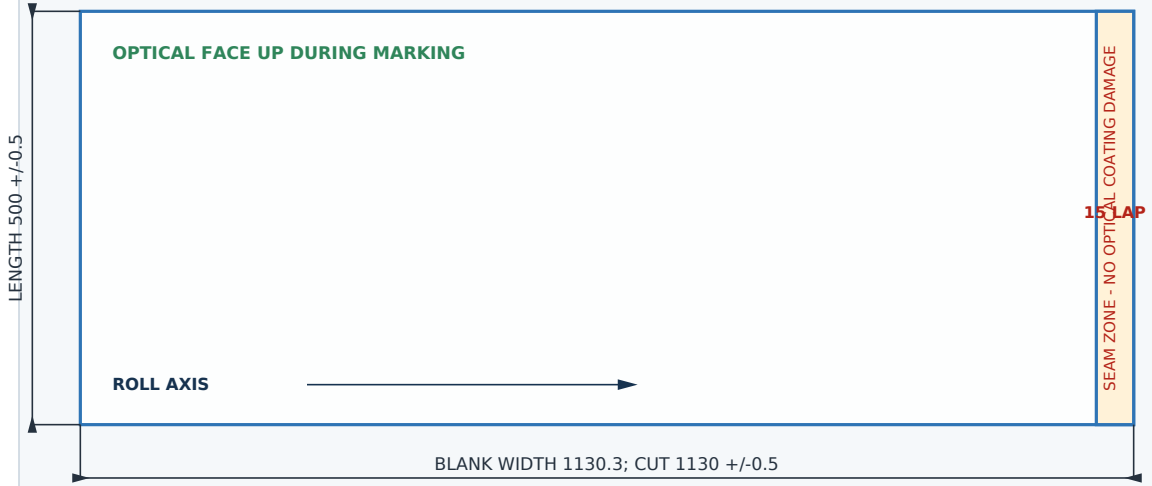
Reflective straight tube module and developed blank

REFLECTIVE STRAIGHT TUBE MODULE

SHEET 2 / 6

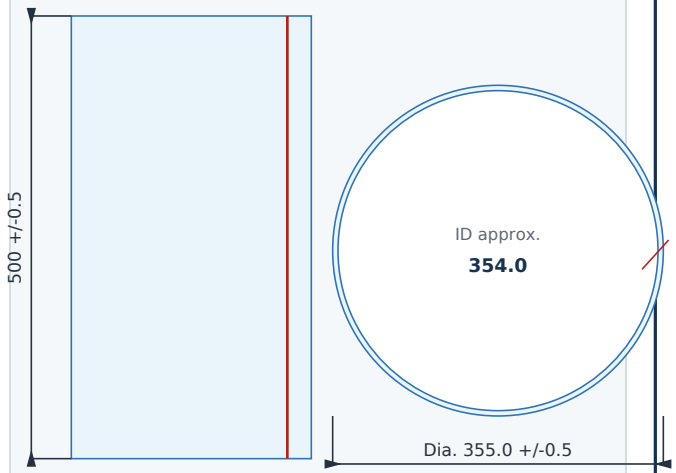
PRELIMINARY FABRICATION DESIGN - ALL DIMENSIONS mm - DRAWINGS NTS

B1 - Developed blank



Material: enhanced specular aluminum, 0.50 mm. Total light reflectance target 98 percent. Keep protective film on during cutting and rolling; remove only after the tube is round and all burrs are removed.

B2 - Rolled module and tolerances



- End squareness <= 0.50
- Roundness: max - min dia. <= 1.0
- No dents > 0.30 deep
- Inside seam step <= 0.30
- Lap edge faces toward diffuser
- No rivet heads in optical bore

This sheet gives the flat blank, formed diameter, seam direction, and inspection tolerances for each repeatable module.

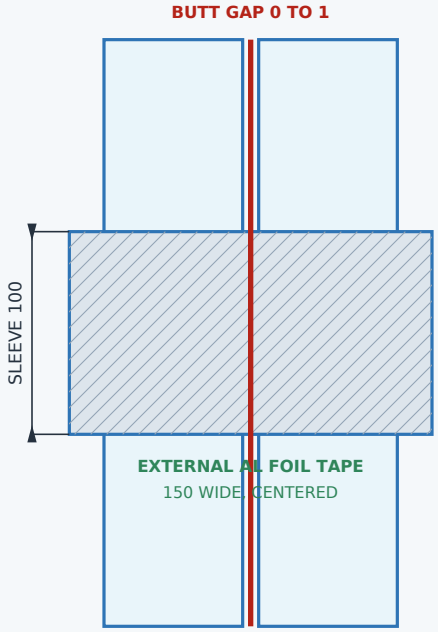
External joints and dome/plenum socket interfaces

JOINTS, DOME INTERFACE, AND LOWER SOCKET

SHEET 3 / 6

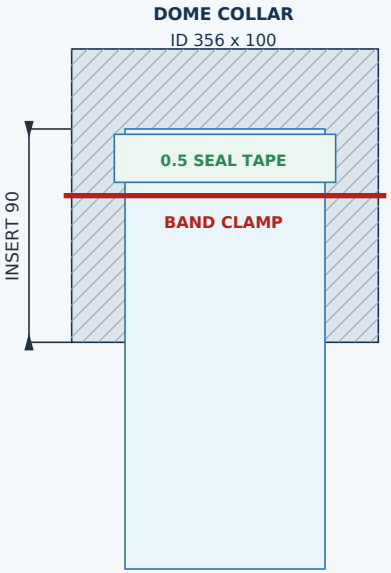
PRELIMINARY FABRICATION DESIGN - ALL DIMENSIONS mm - DRAWINGS NTS

C1 - Straight butt joint



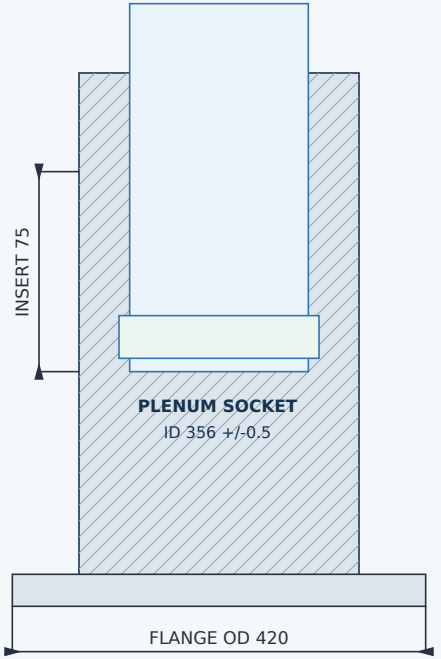
Split sleeve: 5052-H32 aluminum, 1.0 thick x 100 wide. Roll to 356.0 bore. Developed strip 1112 x 100, leaving 8 to 12 mm clamp gap. Add two 30 x 100 x 1.5 clamp ears and two M5 bolts.

C2 - Existing dome collar interface



Keep 10 mm clearance below the collar top. Seal with compatible 0.5 x 25 butyl/EPDM tape and an external stainless band clamp. Do not drill the optical bore.

C3 - Ceiling plenum socket



Seal with 0.5 x 25 compatible tape. Secure using an external band clamp. Support the plenum from framing; the diffuser trim is not structural.

All connections remain external to the optical bore. The tube ends butt together and are sealed outside the light path.

Optional fixed 30-degree elbow and structural supports

OPTIONAL 30-DEG ELBOW AND STRUCTURAL SUPPORTS

SHEET 4 / 6

PRELIMINARY FABRICATION DESIGN - ALL DIMENSIONS mm - DRAWINGS NTS

D1 - Fixed 30-deg miter elbow

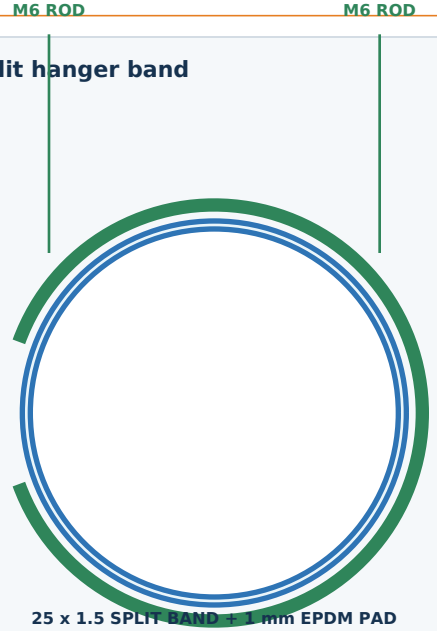


MITER TEMPLATE OFFSET FROM CENTERLINE

Angle	Offset
180 deg	**-47.6 mm**
150 deg	**-41.2 mm**
120 deg	**-23.8 mm**
90 deg	**+0.0 mm**
60 deg	**+23.8 mm**
30 deg	**+41.2 mm**
0 deg	**+47.6 mm**

Cut both tube ends at 15 deg and mirror them. Across a 355 mm tube, the long-to-short cut difference is 95.1 mm. Butt the edges with gap <= 1.5 mm. Bridge externally with four 20 x 120 x 0.8 straps at 90 deg, then seal with 100 mm aluminum foil tape. This fixed elbow is optional; every bend reduces delivered light.

D2 - Split hanger band



Support spacing: <= 1000 mm for the prototype and within 200 mm of each elbow, roof transition, and ceiling plenum. Attach rods only to verified structure. Do not hang the tube from the dome, curb cap, drywall, or diffuser ring. Allow axial thermal movement at one support in each run.

Use the elbow only where obstruction avoidance is necessary. Keep the run as straight and short as practical for best light delivery.

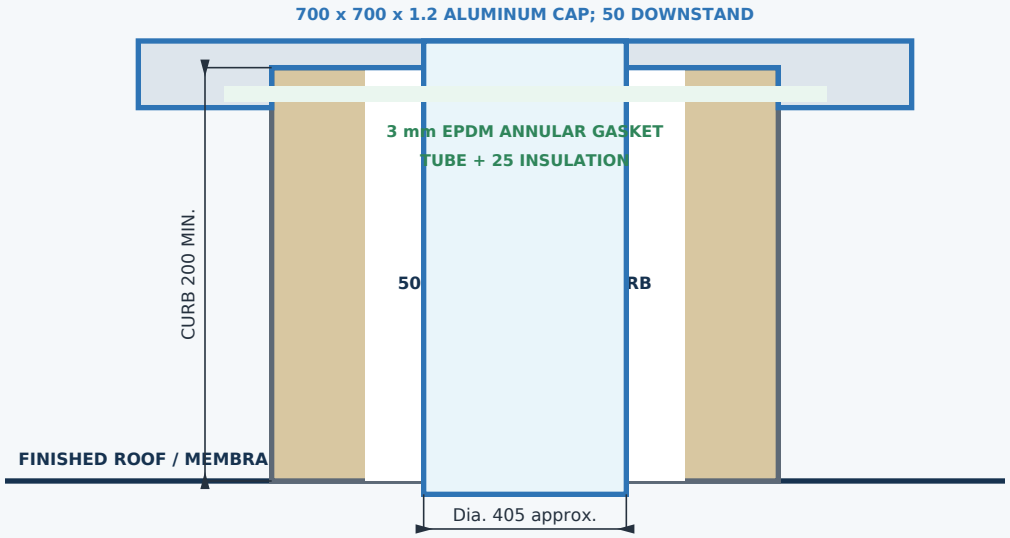
Generic curb-mounted roof adapter

GENERIC CURB-MOUNTED ROOF INTERFACE

SHEET 5 / 6

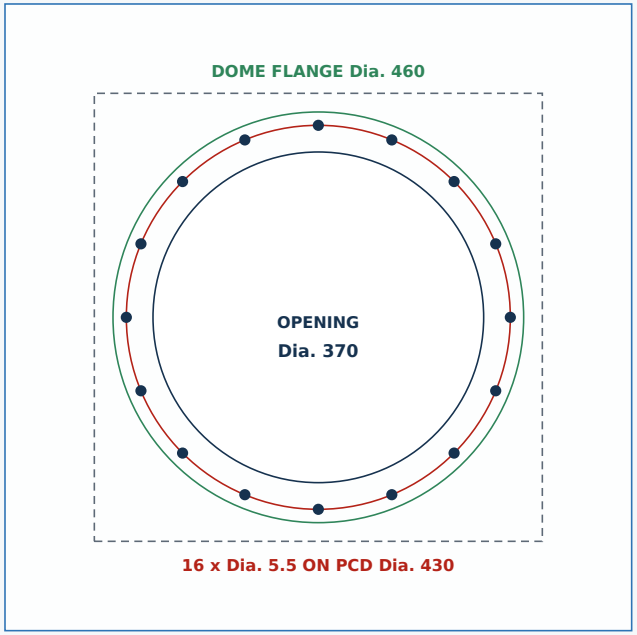
PRELIMINARY FABRICATION DESIGN - ALL DIMENSIONS mm - DRAWINGS NTS

E1 - Section through curb



The cap is a dome mounting adapter, not a universal flashing. The roofing contractor must integrate membrane, shingles, metal panels, drainage, ice/water protection, curb attachment, and counterflashing for the actual roof system.

E2 - Cap plate plan



CAP PLATE 700 x 700

Cap central flat zone Dia. 500 minimum. Opening Dia. 370. Backing ring below cap: OD 456, ID 374, 3.0 aluminum, 16 holes Dia. 5.5 on PCD Dia. 430.

The cap plate provides the dimensional interface to the existing dome. Site-specific curb, flashing, and attachment remain the responsibility of qualified building professionals.

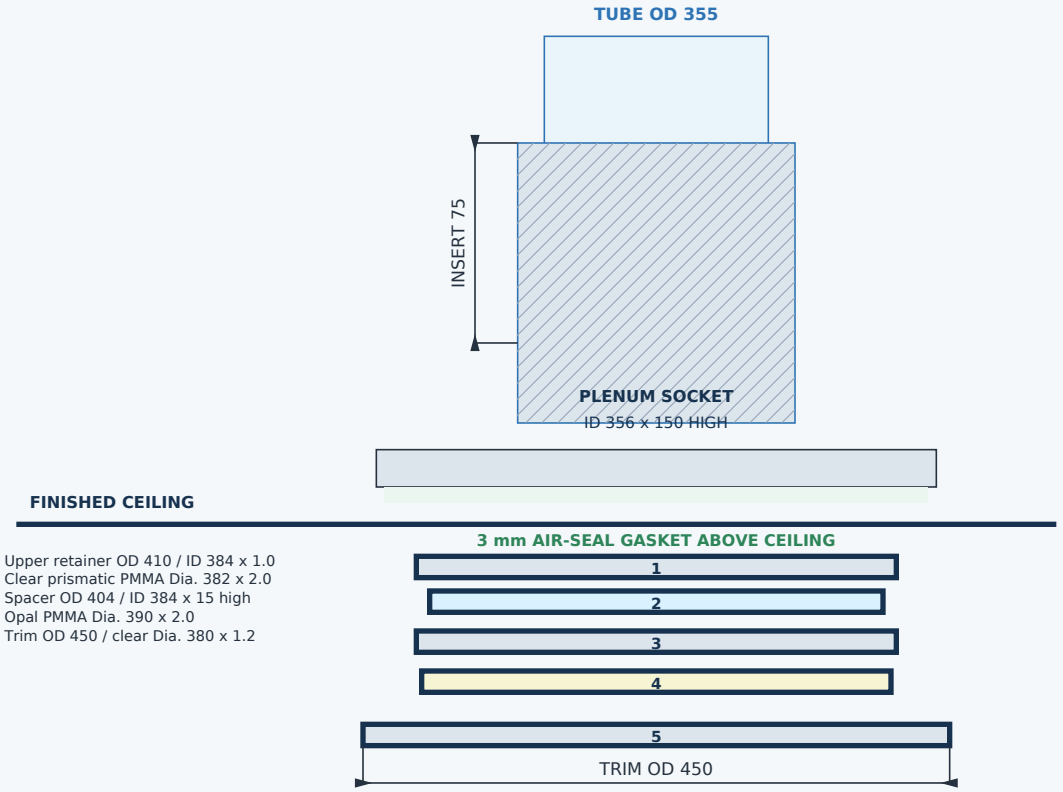
Ceiling plenum, air seal, trim ring, and diffuser

CEILING PLENUM, AIR SEAL, TRIM, AND DUAL DIFFUSER

SHEET 6 / 6

PRELIMINARY FABRICATION DESIGN - ALL DIMENSIONS mm - DRAWINGS NTS

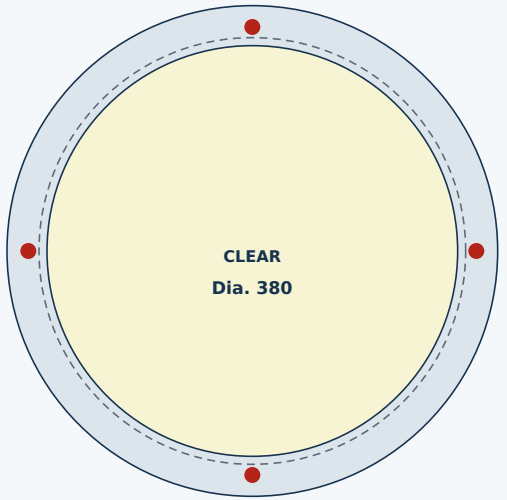
F1 - Exploded section



1 Upper retainer OD 410 / ID 384 x 1.0
2 Clear prismatic PMMA Dia. 382 x 2.0
3 Spacer OD 404 / ID 384 x 15 high
4 Opal PMMA Dia. 390 x 2.0
5 Trim OD 450 / clear Dia. 380 x 1.2

F2 - Plan and service dimensions

CEILING ROUGH OPENING Dia. 400 +/-2



4 SPRING CLIPS AT 90 deg

If exact diffuser transmission is unavailable, prototype 60, 70, and 80 percent opal sheets and choose the lowest-glare option that still meets the room illuminance target.

The diffuser is removable from below for cleaning and optical trials. The plenum must be supported independently from the ceiling finish.

1. Bill of materials and cut list

ITEM	PART	MATERIAL	QTY	CUT / FINISHED SIZE	CRITICAL REQUIREMENT
T1-T4	Straight tube	MIRO-SILVER 27 or equal enhanced specular Al	4	Blank 1130 x 500 x 0.50; formed Dia. 355	Optical face inward; total reflection target 98 percent
J1-J3	Joint sleeve	5052-H32 Al	3	1112 x 100 x 1.0; roll to Dia. 356 bore	8 to 12 clamp gap; no internal hardware
J4	Dome seal	Compatible butyl/EPDM tape	1	0.5 x 25 x 1120	Continuous, compressed; exterior only
J5	Foil joint tape	Outdoor aluminum foil tape	4	150 wide x approx. 1120 each	Seal each butt joint and lower socket
H1-H3	Hanger band	1.5 Al or stainless + EPDM pad	3	25 wide; roll to tube OD	Two M6 rods; structure verified
R1	Curb cap	5052-H32 Al	1	700 x 700 x 1.2; opening Dia. 370	Central flat zone Dia. 500
R2	Backing ring	5052-H32 Al	1	OD 456 / ID 374 x 3.0	16 holes Dia. 5.5 on PCD Dia. 430
P1	Plenum socket	5052-H32 Al	1	ID 356 x 150 x 0.8; flange OD 420	Tube insertion 75
D1	Upper diffuser	Clear microprismatic PMMA	1	Dia. 382 x 2.0	Deburr and retain mechanically
D2	Lower diffuser	Opal PMMA	1	Dia. 390 x 2.0	Trial 65 to 75 percent transmittance
D3	Trim ring	1.2 Al or code-compliant polymer	1	OD 450 / clear Dia. 380	Four service clips at 90 deg
I1	Tube insulation	Code-compliant closed-cell foam	As req.	25 thick	Continuous exterior vapor seal

MATERIAL CONTROL

Do not substitute ordinary mill-finish aluminum for the optical tube. The design depends on a protected enhanced-specular surface. Alanod MIRO-SILVER 27 is available at 0.40 to 0.50 mm and lists total light reflection of 98 percent; an equivalent material must provide documented optical performance and be formable without surface cracking.

2. Fabrication and assembly sequence

STEP	OPERATION	DETAIL / ACCEPTANCE
1	Measure site	Measure from the existing dome-collar underside to the ceiling-plenum top along the intended centerline. Add 165 mm for interface insertions.
2	Plan modules	Divide required cut length into sections <= 600. Keep joints clear of framing, elbows, and hanger locations.
3	Cut optical sheet	Cut blanks with protective film intact. Use shears or a guarded fine-tooth saw. Deburr the back edge without scratching the optical face.
4	Roll cylinders	Roll with optical face inward to Dia. 355. Form a 15 mm longitudinal lap; keep the inner edge step <= 0.30.
5	Close seam	Use compatible high-strength acrylic transfer tape in the lap and exterior foil tape. Do not put rivet heads inside the optical bore.
6	Make sleeves	Roll each 1112 x 100 x 1.0 strip to a 356 bore. Add clamp ears and M5 bolts; trial-fit without crushing the tube.
7	Build roof adapter	Fabricate the cap and backing ring, then have the site-specific curb, attachment, and flashing reviewed before opening the roof.
8	Build plenum	Roll the Dia. 356 socket, form the OD 420 flange, add two structural brackets, gasket, trim, and removable lenses.
9	Dry assemble	Assemble on the floor. Confirm roundness, bore continuity, joint gaps, diffuser removal, and overall centerline length.
10	Install and support	Install roof interface, tube modules, hangers, insulation and vapor seal, then plenum and diffuser. Remove all protective films last.

Assembly controls

- Orient every longitudinal lap edge toward the diffuser so incoming rays meet the smooth side first.
- Align all modules within 2 mm over the full run; avoid visible steps at joints.
- Tighten split sleeves only until the tube cannot rotate by hand. Crushing or ovalizing the bore reduces performance.
- Seal every exterior tube joint airtight before applying insulation. Continuously seal the insulation vapor layer as required by the building climate and code.
- Keep the tube isolated from copper, wet treated wood, masonry salts, and incompatible sealants using approved separation materials.
- Provide access to remove the diffuser and clean the tube without dismantling the roof interface.

CONDENSATION AND FIRE

The 25 mm insulation shown is a prototype starting point, not a universal energy or condensation solution. Determine insulation thickness, vapor-retarder location, fire rating, smoke development, and penetration detailing for the actual climate, roof assembly, and occupancy.

3. Verification, optical commissioning, and limits

CHECK	METHOD	INITIAL PROTOTYPE TARGET
Tube diameter	Measure at 0, 45, 90, and 135 deg at both ends	355.0 +/-0.5; max-min <= 1.0
End squareness	Flat plate and feeler gauge	<= 0.50 gap
Joint alignment	Straightedge across each sleeve and bore inspection	Axis offset <= 1.0; butt gap 0 to 1.0
Air leakage	Low-pressure smoke or tracer test before insulation	No visible leakage at seams, joints, plenum, or dome collar
Water control	Qualified roof spray test including upslope and side laps	No water inside curb, insulation, ceiling plenum, or room
Support	Inspect rods, brackets, clamps, and thermal movement	No load on dome, flashing, drywall, or diffuser
Optical throughput	Compare outdoor horizontal illuminance with room-side diffuser illuminance or flux	Record clear-day and overcast baselines; no joint shadow bands
Glare and diffusion	View at normal room positions under low and high sun	No exposed bright tube image; change opal transmission if needed
Thermal cycle	Minimum 10 hot/cold cycles with inspection	No tape release, condensation staining, corrosion, distortion, or seal movement

Optical configuration rules

- Keep the run straight whenever possible. Use the optional 30-deg elbow only to clear an obstruction and avoid more than two elbows.
- Keep tube interior dry, dust-free, and free of fingerprints. Clean only with the optical-material supplier's approved method.
- Do not paint the optical surface or cover it with ordinary aluminum tape except for a narrow repair approved during prototype testing.
- Longer runs are possible, but the prototype should be photometrically tested at the actual length and bend count before the design is released.

Primary technical references

- VELUX USA, Pitched TMR rigid Sun Tunnel product data: 14-inch system, >98 percent total reflectance, 45-deg adjustable elbow benchmark, and installation-length information. https://pro.veluxusa.com/products/sun-tunnels/rigid-sun-tunnel/tmr

- Alanod, MIRO-SILVER 27 product data sheet, dated 2025-03-03: 0.20 to 0.50 mm material and 98 percent total light reflection. https://alanod.com/_Resources/Persistent/7/2/f/e/72feb130d99a2dd1b4dbf3f2fe603ca72935d5a2/MIRO-SILVER%2027-EN.pdf

- Solatube SolaMaster specification: coordinate roof flashing with insulation, air barriers, vapor retarders, and weather-tight installation. https://solatube.com/wp-content/uploads/2021/02/Arc_Spec_08625_SolaMaster_750_DS.pdf

- PLEXIGLAS Satinice product information: matte acrylic diffusion options for lighting applications. https://www.plexiglas.de/en/products/plexiglas/plexiglas-satinice

- VELUX USA skylight testing information: structural and performance testing context for commercial roof products. https://pro.veluxusa.com/tools/architects/skylight-testing-data

RELEASE AUTHORITY

This package is suitable for prototype fabrication and optical development. It is not a certified skylight, roofing, firestop, structural, or energy-code design. Permanent installation requires review and approval by qualified professionals and any required laboratory or authority having jurisdiction.